What Exists Beyond the Sum-of-Sources Formulation?

ARC White-Box Estimation Challenge 2026 - Phase 2

Alternative formulations for 1024 x 16 bias-free ReLU MLPs
Target lens: raw final-layer MSE ~2e-8 with compute <= 0.16 B

One-sentence answer: No - sum-of-sources is not the only formulation. Public evidence has ruled out many specific implementations, but at least 10 mathematically distinct state representations remain logically open. The four most interesting are: (i) quenched TAP/AMP/cavity, (ii) gate-field plus amplitude models, (iii) operator/kernel-mean embeddings, and (iv) input-function TT/PCE compression.

Research scope

Excluded on purpose: plain Monte Carlo/QMC, ordinary multilevel MC, standard K2/K3/K4 source-bank propagation, finite-width 1/n diagrammatics, ensemble-only DMFT/replica, direct learned weights->means models, and the already-tested full kappa4-memory chain.

1. Executive summary

- Strong conclusion: Saying “all alternatives are dead” is stronger than the evidence supports. Public WHest logs mostly rule out specific implementations; there are very few family-level impossibility theorems.
- Highest-upside non-source-bank route: TAP/AMP/Expectation-Propagation style scalar messages on the actual weight matrices, with Onsager/cavity corrections. This is not ensemble DMFT; the algorithm operates on the realized weight matrices W.
- Most mechanistically targeted route: binary gate field (dichotomized-Gaussian / pairwise max-entropy surrogate) plus positive-amplitude marks. This directly models the sign-switching structure that drives ReLU error.
- Most “different coordinate system” route: approximate the entire input-to-output function in a compressed tensor/spectral basis (TT-PCE / functional tensor train), then read the expectation from the degree-zero coefficient.
- Most exact-but-risky route: activation-region / Gaussian-polytope integration. Exact in principle because ReLU network is piecewise affine; 1024-D region count is the barrier.
- Borderline but important: saddlepoint/CGF propagation is not a finite cumulant truncation; it can resum the effects of many cumulants non-polynomially. However, without a dependence model it is likely to fail again.
- No public challenge implementation was found in this search for forward AMP/TAP, gate-field DG, TT-PCE/TT-cross, sparse characteristic-function state, kernel-mean embedding/Koopman propagation, or saddlepoint-CGF. “Not found” does not mean nobody has tried it; it means no public evidence was located.

Most important evidence rule: a failure of a 1D conditional mixture does not rule out adaptive full GMMs; input active-subspace R^2 ~1e-3 does not rule out TT-PCE rank structure; failure of a Stein control variate does not rule out a deterministic Stein-kernel state; and failure of a locally linearized Gaussian copula does not rule out every copula closure.

2. Problem geometry and the budget filter

Official Phase-2 geometry: width n=1024, depth L=16, zero biases, He-Gaussian weights, x~N(0,I), and per-network budget B=2^41 FLOPs. The official metric uses final-layer MSE times max(0.1, C/B).

Engineering lens for this report: the target is C/B <= 0.16. Therefore, a viable method must avoid the source bank’s O(number_of_sources x n^3) growth and remain in O(n^2), low-single-digit O(n^3), or strongly compressed function-space computation.

Decision criterion: being an “interesting” method family is not enough. Every branch must have a cheap premise test that can expose rank explosion, component explosion, or closure bias before a full 1024 x 16 implementation.

3. Public Failures: What Is Actually Closed, and What Is Not

Public result	What the evidence supports	What the evidence does not support
1D conditional Gaussian mixtures failed; score worsened	fixed 1D/2D conditioning + repeated covariance propagation has poor economics	all adaptive mixture propagation is dead

Public result	What the evidence supports	What the evidence does not support
Gaussian scale-mixture closure: ~2.33e-6 raw in 504aldo probe	one scalar/scale latent does not explain the relevant non-Gaussianity	multi-component directional GMM split/merge is dead
input active-subspace linear R^2 ~1e-3 in barnobarno probe	a few global input directions do not linearly explain the final residual	tensor-train separability or local/nonlinear subspaces are dead
shallow Stein controls gave only small variance reduction	those exact-zero-mean input controls do not repay sampling noise	Stein kernel / zero-bias distributional closure is dead
local-linearized copula paper is structurally bad at zero bias	linearization at x=0 sits on every first-layer ReLU kink	all copula models are dead
source compression / Tucker / per-source rank ladders hit cliffs	leg/source-based K3 compression has a serious rank wall	all representations of the distribution have the same wall

4. Formulation map: 14 alternatives beyond ordinary sum-of-sources

# / family	State	Distinctness	Budget plausibility	Research priority
1 TAP/AMP/cavity	scalar/vector messages + Onsager terms	Genuine	Very strong if derivable	A
2 Gate-field DG/Ising + marks	gate probabilities/correlations + positive amplitudes	Genuine	O(n^2)-ish target	A
3 Koopman/KME/Galerkin	finite dictionary of observables / mean embedding	Genuine	K small => strong	A-
4 TT-PCE / input chaos	compressed function coefficients in input coordinates	Genuine axis change	rank-dependent	A-
5 Functional TT / TT-cross integration	black-box function tensor cores	Genuine	rank-dependent	B+
6 EP / polyhedral gate integration	Gaussian mass over gate inequalities	Genuine	few sweeps / selected gates	B+
7 Characteristic-function state	sparse Fourier/CF evaluations	Genuine	frequency-rank dependent	B
8 Copula + rich marginals	marginal CDFs + dependence copula	Genuine	O(n^2)-O(n^3)	B
9 Adaptive structured GMM	few split/merge components	Genuine	only if M small	B

# / family	State	Distinctness	Budget plausibility	Research priority
10 Transport-map closure	compressed map from Gaussian base to activations	Genuine	rank/order dependent	B-
11 Quenched random-matrix DE	resolvents/local laws + actual-W corrections	Genuine if anisotropic	few n^3 ops possible	A-/B+
12 Saddlepoint / CGF	log-MGF values; tail via saddlepoint	Borderline: all-cumulant resummation	cheap if dependence solved	A- hybrid
13 Stein-kernel / zero-bias	Stein discrepancy / transformed law	Borderline	O(n^2) target	B
14 Angular harmonic / heat-semigroup	spectral/semigroup representation	Borderline with chaos/PCE	compression needed	C+

1. Quenched TAP / AMP / cavity message passing

Classification	GENUINELY DISTINCT - HIGH UPSIDE
Core representation	Per-neuron (or small vector) messages: cavity mean/variance, susceptibility, and Onsager reaction terms; no per-birth K3 source bank.
Why it is not sum-of-sources	Forward distribution is represented by local messages on the dense random linear maps, not by a global third-order tensor. TAP/AMP can run on the actual W, unlike an ensemble-only DMFT scalar recursion.
Phase-2 fit	Dense i.i.d. Gaussian matrices are exactly the regime where AMP-style state evolution and Onsager corrections are strongest. If a forward-marginal analogue can be derived, each iteration can be O(L n^2) rather than O(number_of_sources n^3).
Main risk	Existing ML-VAMP/GAMP theory is mostly inverse inference with observations. With no observations, a naive adaptation may collapse to ordinary Gaussian mean-field and lose the finite-n quenched correction we need.
First falsifier	Derive a one-layer cavity/EP update for z=W^T h with h distributed from a teacher-forced previous layer. On 20 MLPs compare one-step final-mean error against K2 and K3. Kill if Onsager/cavity terms give <30% residual reduction or if >4-8 sweeps are needed.

Evidence note: ML-VAMP and ML-BiGAMP show that multilayer dense random nonlinear models admit scalar/state-evolution message passing in large-system limits; no WHest-specific forward estimator was found.

2. Gate-field model: dichotomized Gaussian / max-entropy gates + amplitude marks

Classification	GENUINELY DISTINCT - HIGH UPSIDE
Core representation	Separate h_i = g_i a_i with binary gate g_i=1[z_i>0] and positive amplitude a_i. Track gate rates, pair correlations, and marked conditional amplitudes.

Why it is not sum-of-sources	The state is a binary switching field plus amplitudes, directly modeling ReLU sign structure instead of polynomial cumulants of h.
Phase-2 fit	At layer 1, gates are exactly thresholded jointly Gaussian. A DG surrogate naturally matches binary rates/covariances; a marked version could preserve E[h_i], E[h_i h_j], and gate-amplitude coupling with O(n^2) state.
Main risk	Pairwise gate statistics may be insufficient; partition functions / high-order gate clusters can become hard. A simplistic marked-DG can collapse to covariance propagation.
First falsifier	Use offline MC/high-moment diagnostics on 128-neuron subsets at layers 4/8/12. Fit DG from gate means/pairs, then test triple gate probabilities and one-step E[ReLU(W^T h)]. Kill if marked-DG fails to beat Gaussian/covariance closure by >=20% on one-step residual.

Evidence note: DG models are established for high-dimensional correlated binary fields. Public WHest searches found no gate-field implementation.

3. Finite Koopman / transfer-operator or kernel-mean embedding (KME)

Classification	GENUINELY DISTINCT
Core representation	Represent the distribution only through expectations of a chosen observable dictionary psi_1,...,psi_K (or RKHS mean embedding); propagate observables under each deterministic layer.
Why it is not sum-of-sources	This propagates linear functionals of the law in feature space rather than raw moments/cumulants or samples. A basis can include gate-sensitive nonlinear observables.
Phase-2 fit	If K is tens to a few hundreds, applying W to basis summaries can be O(K n^2). A sparse conditional-mean/Koopman operator gives a natural place to retain exactly the observables that predict E[ReLU].
Main risk	Closure is the whole problem: generic nonlinear maps need an infinite observable space. A poor dictionary becomes another low-order moment method; an over-rich dictionary explodes.
First falsifier	Start with 32-128 analytic observables (CDF/gate, |z|, z_+, z_+^2, selected pair gates) and fit/projection using only public training MLPs or analytic quadrature. On held-out networks test whether the next-layer mean lies in the span with >90% residual reduction. Kill if K must grow with n.

Evidence note: KME/CME uncertainty propagation and Koopman lifting are established in nonlinear dynamical systems, including sparse operators.

4. Input-space polynomial chaos in TT/CP form (TT-PCE)

Classification	GENUINELY DISTINCT AXIS
Core representation	Approximate each target neuron as a function f_j(x) of the original Gaussian input using Hermite/PCE coefficients compressed in tensor-train or canonical low-rank format.
Why it is not sum-of-sources	K3 source propagation compresses an output cumulant tensor across neurons/sources; TT-PCE compresses the input-coordinate dependence of the deterministic network function. Expectation is simply the degree-0 coefficient.

Phase-2 fit	The first layer is a ridge function, which is extremely structured in input space. Tensorized PCE can beat the combinatorial basis size when TT ranks stay small.
Main risk	Dense random W may destroy separability quickly. The chaos spectrum in public WHest probes has a long high-order tail, so low polynomial degree alone is not enough; success requires low tensor rank despite high degree.
First falsifier	On n=64/128, depth 16, run TT-cross/TT-PCE for 8-16 chosen outputs. Measure TT rank vs depth and integral error. Kill if median TT rank >32-64 by layer 4 or projected model evaluations exceed the 0.16B budget.

Evidence note: Low-rank PCE and tensor-train uncertainty quantification are established high-dimensional tools; this exact WHest compression axis appears publicly untested.

5. Functional tensor train / TT-cross direct Gaussian integration

Classification	GENUINELY DISTINCT
Core representation	Treat f_j(x) itself as a tensorized black-box function on product Gaussian quadrature grids; recover low-rank TT cores adaptively, then integrate cores exactly.
Why it is not sum-of-sources	No layerwise law is propagated at all. The algorithm approximates the final deterministic function and integrates it.
Phase-2 fit	TT-cross can require O(d q r^2) function probes rather than q^d when ranks r are small; Gaussian measure is separable.
Main risk	Each function probe is a full 16-layer forward pass. Dense random mixing can make TT ranks huge; then it degenerates into expensive cubature.
First falsifier	Small-width rank-growth probe first. Compare QTT/TT-cross rank for final neurons across depth 2,4,8,16. Kill immediately if ranks scale roughly linearly with width or if cross evaluations exceed matched MC count with no large accuracy gain.

Evidence note: Tensor-train integration and tensor neural-network quadrature have polynomial-in-dimension complexity only under low-rank structure.

6. Activation-region / Gaussian-polytope integration

Classification	GENUINELY DISTINCT - EXACT IN PRINCIPLE
Core representation	A ReLU network partitions input space into polyhedra. On each activation region, every neuron is affine; expectation is a sum of truncated-Gaussian affine moments over regions.
Why it is not sum-of-sources	The state is geometric: activation patterns and Gaussian mass of polytopes, not cumulants.
Phase-2 fit	For shallow/small PL networks there are analytic moment formulas. A probability-mass beam could in principle keep only regions carrying nearly all Gaussian mass.
Main risk	At 1024x16 the region count is the obvious combinatorial wall. Zero bias places the Gaussian center on many hyperplanes, making mass spread especially dangerous.

First falsifier	Estimate activation-pattern mass concentration from offline samples. If top 256/1024 patterns capture only a tiny fraction of mass, kill full-region beam. A more local variant can condition only on the last 1-2 layers and use EP for their gate polyhedra.

Evidence note: Exact/analytic Gaussian moments of piecewise-linear networks exist for limited architectures; general ReLU region geometry is polyhedral and can grow rapidly.

7. Expectation Propagation over gate inequalities / polyhedral cavity

Classification	GENUINELY DISTINCT
Core representation	Represent active/inactive ReLU constraints as truncation factors; iteratively replace them with Gaussian site factors using cavity distributions and moment matching.
Why it is not sum-of-sources	EP works with approximate factors/partition functions and truncated Gaussian geometry rather than carrying third-order source tensors.
Phase-2 fit	Rank-one site updates can be cheap. A block-EP restricted to influential gates could add non-Gaussian gate information on top of a Gaussian backbone without explicit region enumeration.
Main risk	EP can be arbitrarily wrong for general polyhedra; 1024 gates per layer can require too many site updates. ReLU is not just an indicator - active amplitude must also be handled.
First falsifier	At one layer with teacher-forced Gaussian input, compare EP-predicted first/second moments after 32,64,128 selected gate factors to exact MC. Kill if error does not fall monotonically or if full sweep count makes cost >~5u/layer.

Evidence note: Cunningham-Hennig-Lacoste-Julien show EP is accurate for many Gaussian regions but can fail badly on arbitrary polyhedra - so it needs aggressive falsification, not blind trust.

8. Characteristic-function / sparse Fourier propagation

Classification	GENUINELY DISTINCT
Core representation	Track phi(t)=E exp(i t^T h) at selected frequencies, or low-rank marginal/joint characteristic-function features.
Why it is not sum-of-sources	Characteristic functions describe the law in frequency space and do not require finite moments. Linear transforms are exact via phi_{Wh}(t)=phi_h(W^T t).
Phase-2 fit	The linear step is ideal. If only a sparse set of frequencies around directions relevant to the next W rows is needed, a small adaptive frequency bank could capture non-Gaussian shape near the ReLU threshold.
Main risk	Coordinatewise ReLU couples frequencies and high-dimensional inversion is expensive. Full multivariate CF is impossible at n=1024; sparse frequency closure may lose dependence.
First falsifier	Teacher-force one layer and retain q=16,32,64 adaptive frequencies per target. Reconstruct E[z_+] from the CF and compare with K3. Kill if q grows rapidly with depth or if joint-frequency rank is high.

Evidence note: Pilipovsky et al. explicitly propagate distributions of ReLU networks through characteristic functions for probabilistic verification.

9. Gaussian copula + rich nonparametric marginals

Classification	GENUINELY DISTINCT IF MARGINALS ARE RICH
Core representation	Track each neuron marginal by quantile/PDF knots and dependence via a Gaussian or low-rank copula correlation matrix.
Why it is not sum-of-sources	Non-Gaussianity is stored in one-dimensional marginals; dependence is separated into a copula rather than encoded in joint cumulant tensors.
Phase-2 fit	Marginal ReLU is exact (truncate/atom at zero), and O(q n) marginal state is cheap. Pair dependence can reuse an O(n^2) correlation backbone.
Main risk	A local-linearized copula model is poor for zero-bias ReLU because x=0 lies on the kink. Weighted sums under a copula still need an efficient transform; Gaussian copula may miss tail/gate dependence.
First falsifier	Do not test the published local-linearization directly. Instead teacher-force true marginals, fit only the copula, and test next-layer positive-part mean. If even oracle marginals + copula cannot beat K2/K3, kill the family.

Evidence note: 2026 Diamzon-Venturi derives PDF/moment propagation with Gaussian-copula surrogates, but 504aldo correctly noted their linearization assumptions are unfavorable for this challenge. That does not falsify an oracle-marginal copula test.

10. Adaptive structured Gaussian mixture split/merge

Classification	GENUINELY DISTINCT
Core representation	Approximate the activation distribution by M Gaussian components, splitting components where ReLU nonlinearity causes largest Wasserstein distortion and merging redundant ones.
Why it is not sum-of-sources	The state is a multimodal density, not a polynomial moment hierarchy.
Phase-2 fit	WGMprop-style adaptive split/merge is explicitly designed for nonlinear neural-network uncertainty. A structured covariance (diag + rank-r) could keep each component much cheaper than full covariance.
Main risk	1024 simultaneous gates can force component explosion. Public WHest failures show 1D conditional mixtures and a Gaussian scale-mixture are insufficient, but not that adaptive directional GMM is impossible.
First falsifier	Implement only on width 128/256 first. Track number of components needed for one-step and rolled mean error. Kill if M>16-32 by layer 4 or if component count doubles repeatedly.

Evidence note: Monchot et al. (ICML 2023) and earlier work use adaptive GMM split/merge through neural networks; public WHest tests covered much narrower mixture classes.

11. Quenched random-matrix deterministic equivalents + anisotropic correction

Classification	GENUINELY DISTINCT IF SAMPLE-SPECIFIC
Core representation	Use actual-W Gram/resolvent/local-law observables to approximate the finite-width distribution; retain a small set of anisotropic diagonal or leave-one-out corrections.
Why it is not sum-of-sources	Unlike ensemble DMFT, quenched/local-law methods can depend on the realized matrices. Unlike K3 source banks, the state can be a few resolvents, traces, diagonals, or susceptibilities.
Phase-2 fit	Random He matrices are exactly random-matrix objects. Deterministic-equivalent and neural-matrix-law literature suggests many bulk quantities self-average; the unexplained residual may live in a small anisotropic correction.
Main risk	Per-neuron 2e-8 accuracy may depend on fluctuations not captured by bulk spectra. A pure deterministic equivalent would only give a typical/ensemble result and fail the task.
First falsifier	Build a residual regression/falsifier using only actual-W observables: row norms, diagonal powers of Gram matrices, trace moments, approximate resolvent diagonals, leverage/susceptibility. Cross-validate across MLPs. If these explain <20-30% of V29 residual, scalar DE route is unlikely.

Evidence note: Tensor Programs III proves asymptotic freeness; deep random-feature work derives deterministic equivalents. The missing piece is a quenched, per-neuron finite-n correction.

12. Saddlepoint / cumulant-generating-function (CGF) propagation

Classification	BORDERLINE: NOT FINITE-ORDER CUMULANT TRUNCATION
Core representation	Track K(t)=log E[e^{tZ}] at selected t and use a saddlepoint (e.g. Lugannani-Rice / Daniels) to recover tail probabilities or E[Z_+].
Why it is not sum-of-sources	It does not store kappa3 source tensors and does not truncate after kappa3/kappa4; a saddlepoint can effectively resum all cumulants encoded by K(t).
Phase-2 fit	For a weighted sum of independent/weakly dependent terms, CGFs add. If TAP/cavity can produce near-independent effective channels, CGF+saddlepoint becomes especially attractive and cheap.
Main risk	The layer activations are dependent; naive product-CGF ignores the exact thing K3 is fixing. Without a dependence/cluster correction this may revert to Gaussian-quality error.
First falsifier	Teacher-force h samples, compute “oracle marginal CGF + independence” prediction for W^T h, then add pair/Bethe corrections. If oracle-marginal independence is already far above 1e-7 one-step MSE, do not port.

Evidence note: Classical saddlepoint approximations are much more uniform in tails than Edgeworth expansions for sums, but the independence assumption is the key obstacle here.

13. Stein-kernel / zero-bias distributional closure

Classification	BORDERLINE BUT DIFFERENT FROM STEIN CONTROL VARIATES
Core representation	Track a Stein kernel, zero-bias transform, or small discrepancy features that directly correct Gaussian integration-by-parts identities for each preactivation.

Why it is not sum-of-sources	A Stein control variate is a sampling trick; a Stein-kernel state is a deterministic description of how the law deviates from Gaussian and can target E[ReLU(Z)] directly.
Phase-2 fit	Stein theory already gives quantitative Gaussian approximation bounds for deep random Gaussian networks. A scalar/pairwise Stein discrepancy might capture the ReLU mean error with much less state than full K3.
Main risk	No known closed recurrent Stein-kernel update for dependent ReLU layers. It may secretly require the same high-order information as cumulants.
First falsifier	Offline teacher-forced test: regress the exact ReLU-mean residual against candidate Stein observables (density-at-zero proxy, zero-bias displacement, pairwise Stein matrix summaries). Kill if out-of-sample R^2 is low or correction is not stable across depth.

Evidence note: Public WHest Stein-CV failures do not test this deterministic-state idea. Stein-method literature supplies approximation identities/bounds, not a ready competition estimator.

14. Angular harmonics / heat-semigroup / spectral smoothing

Classification	BORDERLINE SPECTRAL FAMILY
Core representation	After exact radial separation, represent the angular function on the sphere via low-degree spherical harmonics/Gegenbauer ridge components; equivalently view Gaussian expectation as a heat-semigroup smoothing of the CPWL function.
Why it is not sum-of-sources	This works in function/spectral space rather than neuron cumulant space. The mean is the degree-0 mode after angular propagation.
Phase-2 fit	Positive homogeneity is unusually favorable: radius separates exactly. ReLU ridge functions have structured spherical harmonic expansions, so a compressed zonal representation is conceivable.
Main risk	The number of harmonics in d=1024 explodes with degree; public chaos-spectrum measurements show a heavy high-order tail. Without tensor compression this is dead, and with low-degree truncation it may become ordinary Hermite moment closure.
First falsifier	On small widths measure angular spectral energy and tensor rank by degree. Kill if degree>6-8 carries material mean correction or coefficient rank explodes. Keep only if a Gram/zonal representation avoids explicit harmonic enumeration.

Evidence note: Useful mainly as a basis-design idea for TT/operator methods, not as a standalone explicit harmonic expansion.

5. The strongest new combinations (not individual methods)

Combination A - TAP/AMP + saddlepoint CGF. Use cavity/Onsager corrections to turn each dense weighted sum into an approximately decoupled scalar channel, then use a saddlepoint approximation rather than a Gaussian/Edgeworth truncation for that scalar channel. If successful, the source bank's dependence correction moves into message passing, while non-Gaussian marginal shape is handled by the CGF.

Combination B - Gate field + marked amplitudes + EP DG/Ising-style gate correlations capture sign structure; EP

handles selected high-influence gate constraints; conditional positive amplitudes carry magnitude. This attacks the ReLU kink directly rather than reconstructing a global K3 tensor.

Combination C - Angular radial separation + TT-PCE/TT-cross Radius is analytically integrated once (already known). The remaining sphere/input-direction function is compressed in TT/PCE. This avoids wasting tensor rank on a nuisance radial dimension and is much more interesting than plain low-degree PCE.

Combination D - Quenched RMT observables as an operator dictionary Use resolvent/Gram/leverage/susceptibility features from actual W as the low-dimensional Koopman/KME dictionary. This is a principled way to inject network-specific finite-width information without source history.

6. What I would test first: a five-gate research ladder

Gate 1 - TAP/AMP feasibility (highest information value): Before full derivation, test whether an Onsager/cavity correction computed from actual W and teacher-forced previous-layer statistics predicts the one-step residual direction. Use 20 public MLPs, layers 4/8/12/15. Promotion: >=30% reduction in one-step mean MSE on held-out MLPs with O(n^2) or <=2u equivalent extra work.

Gate 2 - Marked gate-field oracle: Give the method oracle gate means/pair correlations from MC and test whether DG + conditional amplitude reconstructs the next-layer mean. This separates "representation insufficient" from "parameter estimation insufficient." Promotion: clear gain over K2/K3 one-step closure; reject if oracle pairwise gate state is still weak.

Gate 3 - TT rank-growth probe: At n=64/128, approximate 8 final neurons with TT-cross/TT-PCE and record TT rank by depth. Promotion: ranks stay <=32-ish and integral error decays faster than matched forward evaluations. Reject on early rank explosion.

Gate 4 - CGF/saddlepoint oracle-dependence test: From teacher samples compute exact univariate marginal CGFs, first ignoring dependence and then adding pairwise/Bethe corrections. If even oracle marginals cannot produce an accurate weighted-sum positive-part mean, the saddlepoint branch is closed.

Gate 5 - sparse operator dictionary: Build a 32/64/128-feature deterministic dictionary mixing gate probabilities, marginal transforms, Gram/resolvent diagonals, and angular features. Fit/projection only on train MLPs; evaluate full rollout on untouched MLPs. Promotion only if error reduction survives without K growing with n.

7. Which routes could realistically hit <=0.16B?

Route	Compute shape if successful	Accuracy ceiling risk	Verdict before experiments
TAP/AMP/cavity	O(iter x L n^2) to low n^3	may collapse to mean-field	Best theoretical gamble
Gate-field + marked amplitude	O(L n^2) state; pair updates	pairwise gates may miss clusters	Best mechanism-targeted gamble
TT-PCE / TT-cross	O(d q r^2) probes/ops if rank low	rank explosion	Best representation gamble

Route	Compute shape if successful	Accuracy ceiling risk	Verdict before experiments
Sparse operator/KME	O(K L n^2) if K small	dictionary closure	Very testable
Saddlepoint-CGF hybrid	cheap after decoupling	dependence is hard	Good add-on, not standalone
Copula + marginals	O(n^2)-few n^3	copula bias	Medium
Adaptive GMM	M x covariance cost	component explosion	Medium-low
Full region/polytope	combinatorial	region explosion	Low standalone

8. Bottom line

Sum-of-sources is not the only possible formulation. It is a very effective representation for the current K3 architecture, but the same expectation problem can be attacked in entirely different coordinate systems: probability-law space, gate geometry, observable/operator space, Fourier space, input-function space, message space, or a quenched random-matrix state.

The main lesson, however, is that naming an alternative family is not a breakthrough. The breakthrough would be a representation whose state dimension does not grow with depth like the source count, while still retaining fixed-W finite-width corrections. Under that criterion, TAP/AMP, marked gate fields, sparse operator/KME states, and TT function compression are the cleanest new bets.

Given the current evidence, my research order is: (1) TAP/AMP one-step derivation and falsifier; (2) marked gate-field oracle; (3) TT rank probe; (4) CGF-saddlepoint hybrid; and (5) sparse operator dictionary. If these fail, then it is reasonable to move to the copula, GMM, and polyhedral branches.

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Research caveat: The report separates published facts from proposed WHest adaptations. AMP/TAP forward propagation, marked gate-field, TT-PCE at 1024x16, and quenched DE corrections are research hypotheses here, not measured competition results. “No public implementation found” is not evidence of nonexistence.